

LARAVEL AFTER DEPLOY

ARCHITECTURE, PERFORMANCE,
AND OPERATIONS AT SCALE

Nuruzzaman Milon



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To my family, for making every effort worthwhile.

Preface

Who this book is for

This book is written for a mid-to-senior Laravel engineer who already knows how to build a Laravel application and is now being asked to keep one alive - the person a growing team hands the architecture decisions, the performance incident, and the migration that has to happen without a maintenance window. It assumes you already know how routing, Eloquent, queues, and testing work in Laravel; it doesn't stop to explain the framework. Every page is instead about the layer of decisions that sits on top of it: how you structure a codebase as more than one team starts touching it, how you keep it fast once traffic outgrows what fits comfortably on one database, how you keep it correct when the same request can arrive twice, how you keep it available when a dependency you don't control goes down, and how you run all of that at 3 a.m. on a Tuesday without it becoming a story you're still telling a year later.

If you're looking for a Laravel tutorial - how to define a route, how to write a migration, how Eloquent relationships work - this isn't that book, and there are excellent ones already. If you're the engineer who's outgrown those books and is now staring at a monorepo, a queue backlog, or a balance that doesn't add up, keep reading.

How to read it

The book is organized into eight parts, and the parts are ordered by dependency rather than importance, so reading front to back works, but it isn't the only sane way through it. Part 1 (Foundations) and Part 2 (Performance Engineering) build vocabulary that later parts assume - "shared kernel," "N+1," "backfill," and so on. Part 5 (Observability & Monitoring) is worth reading earlier than its page number suggests, because "correlation ID," "SLI/SLO/SLA," and "burn rate" get reused heavily in Parts 4, 6, and 7. Beyond that, Parts 3 through 8 are largely independent of each other, so if you picked this book up because of a live incident with your queues, your database failover, or your CI pipeline, it's reasonable to skip straight to the relevant chapter. Chapter 34, the capstone, assumes you've read - or at least skimmed - everything before it: it's a single worked incident that touches nearly every decision in the book, and it's meant to be read last.

Each chapter follows the same shape. It opens with a concrete production failure - drawn from more than thirteen years of real incidents, anonymized - then works through the pattern that would have prevented or fixed it, and closes with a short "Chapter recap" you can scan later without rereading the whole chapter. Not every one of those incidents happened in a Laravel project; over those years I've worked across many languages and frameworks, and where the original system wasn't Laravel, I've translated the failure into a Laravel setting so it can be demonstrated in the stack this book is about.

About the examples

Every incident described in this book actually happened in real production systems. The failure modes, the trade-offs, and the fixes come from systems I helped build and then had to keep running, not from thought experiments. What isn't real is any identifying detail: company names, product names, team names, exact numbers, and anything else that could point back to a specific employer or client has been changed, generalized, or recombined. The named fictional businesses that recur throughout the book - Northfield (an order-processing platform), Vaultly (a wallet service), Reelcast (a livestreaming and short-video app), and others - are stand-ins for this kind of company, not case studies of any single one; they exist because those are the systems that reliably produce the failure modes worth discussing, not because they map one-to-one onto any company you could name. Nothing here discloses a trade secret, a proprietary architecture, or confidential information belonging to any organization I've worked with; the anonymization exists specifically to protect that while keeping the underlying lesson intact. The architecture decisions and opinions in these pages are my own.

A note on unfamiliar terms

Abbreviations are spelled out the first time they're used in a chapter, but if you're dropping into a chapter out of order, Appendix A (Glossary) collects the recurring ones - SLO, SLA, SLI, MTTR, RPO/RTO, ADR, and the rest - in one place, each with a pointer back to the chapter that covers it properly.

Chapter 8 - Safe Schema Evolution at Scale

The migration that only failed in production

On Northfield, a fictional order-processing platform, a routine cleanup task renames a column: `orders.buyer_email` becomes `orders.customer_email`, to match naming used everywhere else in the domain. The migration is small, obviously correct, and reviewed without objection. In staging, against a ten-thousand-row table, it runs in forty milliseconds. In production, against a table with two hundred million rows, the `ALTER TABLE` takes an exclusive lock for twenty-five minutes. Every write to `orders` blocks behind it. The API starts returning timeouts, the on-call engineer gets paged, and the migration that "obviously" worked has to be killed mid-run, leaving the schema in an ambiguous state.

Nothing about the migration was wrong, syntactically. It was tested at the wrong scale, the same failure mode as Chapter 7, applied to schema changes instead of queries. This chapter covers the patterns that make schema changes safe regardless of table size: non-blocking DDL, a backward-compatible rename pattern, batched backfills, and - the part teams forget most often - tracking who else reads your schema before you assume a "safe" change is actually safe for everyone.

Why "fast in staging" doesn't transfer

The cost of a schema change is a function of data volume, not query complexity, and that cost varies by database engine and by the specific operation:

- Adding a nullable column with no default is metadata-only and near-instant on both MySQL (InnoDB, modern versions) and PostgreSQL, regardless of table size.
- Adding a column with a default value, or adding a **NOT NULL** constraint to an existing column, historically required rewriting every row to populate or validate the new value - a cost proportional to table size. Recent versions of both engines have narrowed this (PostgreSQL avoids a full rewrite for constant defaults since version 11; MySQL's behavior depends on the specific **ALGORITHM** used), but the safe assumption is: verify the specific operation against the specific engine version in use, on a copy of production-sized data, before trusting it in staging.
- Renaming a column, dropping a column, or changing a column's type can each range from instant metadata changes to full table rewrites, again depending on engine and version.

The rule that survives all of that engine-specific detail: never trust a migration's timing from a small staging database. Run it against a production-sized copy (a recent snapshot, restored somewhere disposable) before it ships, and know in advance whether the specific operation takes a blocking lock.

Concretely, adding a **NOT NULL** constraint to **orders.customer_email** on PostgreSQL as a single step forces the database to scan and validate every one of two hundred million rows while holding a lock that blocks writes for the

duration:

```
/*  
 * Locks orders for the full validation scan - the whole table,  
 * all at once.  
 */  
Schema::table('orders', function (Blueprint $table) {  
    $table->string('customer_email')->nullable(false)->change();  
});
```

Splitting it into "add the constraint without enforcing it yet," "validate it separately," and only then "make the column formally **NOT NULL**" turns one long exclusive lock into a near-instant metadata change plus a scan that only takes a brief lock at the very end:

```
-- Step 1: instant - records the constraint without checking
existing rows.
ALTER TABLE orders ADD CONSTRAINT customer_email_not_null
    CHECK (customer_email IS NOT NULL) NOT VALID;

-- Step 2: scans the table, but only takes a brief lock to flip
the
-- constraint's state once the scan finds no violations - not for
the
-- duration of the scan itself.
ALTER TABLE orders VALIDATE CONSTRAINT customer_email_not_null;

-- Step 3: after the validated check proves there are no nulls,
PostgreSQL
-- can set the column attribute without repeating the full-table
scan.
ALTER TABLE orders
    ALTER COLUMN customer_email SET NOT NULL;
```

Run as separate deploys, the first one is safe to ship immediately; validation and the final **SET NOT NULL** can run once the backfill from the next section has finished, with no code depending on their timing.

Concurrent and non-blocking index creation

Adding an index to a large table is one of the most common migrations to get wrong, because a plain **CREATE INDEX** takes a lock that blocks writes for as long as the index build takes - which, on a two-hundred-million-row table, can be tens of minutes.

PostgreSQL's answer is `CREATE INDEX CONCURRENTLY`, which builds the index without holding a write lock, at the cost of a slower two-pass build and a real failure mode: if it's interrupted, it can leave behind an invalid index that has to be dropped and retried, not resumed.

```
CREATE INDEX CONCURRENTLY idx_orders_customer_created
ON orders (customer_id, created_at);
```

MySQL's InnoDB engine supports online DDL for many index operations via `ALGORITHM=INPLACE, LOCK=NONE`, which allows concurrent reads and writes during the build - but not every DDL operation supports every algorithm, and the safe habit is to check the specific operation against the running engine version rather than assume.

```
ALTER TABLE orders
ADD INDEX idx_orders_customer_created (customer_id,
created_at),
ALGORITHM=INPLACE, LOCK=NONE;
```

Laravel's migration files don't manage this distinction automatically - a `Schema::table()` migration using `$table->index()` issues a plain `CREATE INDEX/ADD INDEX` unless the raw SQL or a database-specific driver option is used to request the non-blocking variant. On a large table, that's a decision to make explicitly, not a default to trust:

```
public function up(): void
{
    /*
     * Schema::table()→index() takes a blocking lock for the
     * build - fine on a small table, a production incident on a
     * two-hundred-million-row one.
     */
    DB::statement(
        'CREATE INDEX CONCURRENTLY idx_orders_customer_created'
        .' ON orders (customer_id, created_at)'
    );
}
```

CREATE INDEX CONCURRENTLY can't run inside the transaction Laravel wraps each migration in by default, so the migration class also needs **public \$withinTransaction = false;** - forgetting that is the most common way this pattern fails silently in a migration file, even though it works fine when run by hand. And because an interrupted concurrent build leaves an unusable index behind rather than rolling back, always check for one before retrying:

```
-- Finds indexes left in an invalid state by an interrupted
CONCURRENTLY build.
SELECT indexrelid::regclass AS index_name
FROM pg_index
WHERE indisvalid = false;

-- Safe to drop and retry - an invalid index is never used by the
planner.
DROP INDEX CONCURRENTLY idx_orders_customer_created;
```

The backward-compatible column rename pattern

Renaming a column directly - `ALTER TABLE orders RENAME COLUMN buyer_email TO customer_email` - is fast as a pure metadata operation on most engines. The real danger isn't the `ALTER` itself; it's that the moment it commits, every piece of code still referencing the old column name breaks atomically, everywhere, at once - including code you don't control (see the next section). The fix is to never do a rename as a single step. Expand it into phases that can each be shipped, verified, and rolled back independently:

Phase	Schema state	Application state	Rollback if something's wrong
1. Add	New column exists, nullable	Not yet used	Drop the new column - no impact

Phase	Schema state	Application state	Rollback if something's wrong
2. Dual-write	Both columns exist	App writes to both columns on every write path; reads still from the old column	Stop dual-writing - old column is still authoritative
3. Backfill	Both columns exist	Batch job populates the new column for existing rows	Backfill is idempotent - safe to re-run or pause
4. Migrate reads	Both columns exist	App reads from the new column; old column no longer read	Revert the read change - old column is still fully populated
5. Stop dual-write	Both columns exist	App writes only to the new column	Revert to dual-writing if any consumer still expects the old column
6. Drop	Old column removed	Fully migrated	Not reversible past this point - this phase only ships once nothing reads the old column

Each phase is a separate, independently deployable change. If phase 4 reveals a bug (the new column has a subtly different meaning than assumed), the rollback is "revert one deploy," not "recover from a completed rename." This is the same expand-and-contract shape used for the shared-package upgrades in Chapter 6 - the pattern generalizes to any change where "old" and "new" need to coexist for a transition window.

Phase 1 is the only phase that touches the database schema directly - it's a plain, additive migration:

```
public function up(): void
{
    Schema::table('orders', function (Blueprint $table) {
        // after() is MySQL-only; omit it on PostgreSQL.
        $table->string(
            'customer_email'
        )->nullable()->after('buyer_email');
    });
}
```

Phase 2's dual-write lives in the model, not in a migration, and is the one line of application code every write path now depends on until phase 5 removes it - with the caveat that "every write path" means every write path that goes through a model instance. `Order::query()->update( ... )` and anything built on `DB::table('orders')` skip model events entirely, so grep for those before trusting this hook, because phase 4's correctness rests on it:

```
protected static function booted(): void
{
    static::saving(function (Order $order) {
        if ($order->isDirty('buyer_email')) {
            $order->customer_email = $order->buyer_email;
        }
    });
}
```

Phase 6's drop is deliberately its own migration, shipped only after the checklist at the end of this chapter is fully checked off:

```
public function up(): void
{
    Schema::table('orders', function (Blueprint $table) {
        $table->dropColumn('buyer_email');
    });
}
```

Batching large backfills

Phase 3 above - populating the new column for two hundred million existing rows - is itself a hazard if done as one statement:

```
// Don't do this on a large table.
DB::table('orders')
  →update(['customer_email' => DB::raw('buyer_email')]);
```

A single **UPDATE** touching every row holds row locks (and, depending on engine and isolation level, can hold them for the duration of one very long transaction), competes with live traffic for the same rows, and - on replicated setups - can create significant replication lag, since a replica applies the same enormous write as one unit. The safe alternative is chunking: touch a bounded number of rows per statement, with a small pause between batches to let replicas catch up and to leave room for live traffic.

```
Order::query()
  →whereNull('customer_email')
  →orderBy('id')
  →chunkById(1000, function ($orders) {
    Order::query()
      →whereIn('id', $orders→pluck('id'))
      →update(
        ['customer_email' => DB::raw('buyer_email')]
      );

    // 100ms - bound replication lag and lock contention
    usleep(100_000);
  });
```

Running this as a resumable Artisan command that processes one bounded batch (tracking the last processed ID) rather than a single migration step

means it can be paused, monitored, and restarted without redoing completed work - important for a backfill that might run for hours against a genuinely large table:

```
final class BackfillCustomerEmail extends Command
{
  protected $signature =
    'orders:backfill-customer-email'
    .' {--fresh} {--dry-run} {--batch=1000}';

  public function handle(): int
  {
    $lastId = $this->option(
      'fresh'
    ) ? 0 : (int) DB::table('backfill_progress')
      ->where('name', 'customer_email')
      ->value('last_id');

    $dryRun = $this->option('dry-run');
    $orders = Order::query()
      ->whereNull('customer_email')
      ->where('id', '>', $lastId)
      ->orderBy('id')
      ->limit((int) $this->option('batch'))
      ->get(['id']);

    if ($orders->isEmpty()) {
      $this->components->info(
        'Backfill complete: no null emails left.'
      );

      return self::SUCCESS;
    }
  }
}
```

```
}

$nextLastId = $orders->last()->id;

if (! $dryRun) {
  /*
   * whereNull again, so a concurrent write that already
   * populated one of these rows isn't overwritten by a
   * stale read.
   */
  Order::query()
    ->whereIn('id', $orders->pluck('id'))
    ->whereNull('customer_email')
    ->update(
      ['customer_email' => DB::raw('buyer_email')]
    );

  DB::table('backfill_progress')->updateOrInsert(
    ['name' => 'customer_email'],
    ['last_id' => $nextLastId],
  );
}

$this->components->info(
  ($dryRun ? '[dry run] ' : '')
  ."Processed {$orders->count()} rows"
  ." (last ID: {$nextLastId})."
);

return self::SUCCESS;
}
}
```

Dispatched via `Schedule :: command('orders:backfill-customer-email')->everyMinute()->withoutOverlapping()`, this makes progress in small, bounded increments and survives being interrupted by a deploy at any point. The cursor lives in a table rather than in the cache for a boring reason that bites people anyway: a cache is allowed to lose your key. An eviction under `maxmemory`, a `cache:clear` during a release, a Redis restart without persistence - any of those silently resets the backfill to ID 0, and the `whereNull` guard means it will quietly grind back through two hundred million already-completed rows without ever reporting that anything went wrong.

`--dry-run` is worth adding to every backfill command before it ever touches a large table: it walks the exact same query and batch-selection logic, reports how many rows and which ID range the next batch would affect, and writes nothing - so a mistake in the `WHERE` clause shows up as a suspicious row count in dry-run output, not as two hundred million incorrectly updated rows. And it's always worth emitting a progress line per batch, not just a single message at the end - a backfill that runs for hours with no output is indistinguishable from one that's silently hung, and the per-batch line (row count, last ID reached) is what turns "is this still running?" into a question the terminal already answers.

Downstream consumers: your schema is a public API

The phased rename above protects the application. It does nothing for anything else that reads the same database directly and isn't part of the deploy you control: a nightly analytics export, another internal service with direct read access to the same tables (an anti-pattern, but a common inherited one), or a BI tool querying a read replica. From their perspective, dropping

`buyer_email` in phase 6 is a breaking change shipped with zero notice, because nothing about the application's own migration process is aware they exist.

The fix starts with an inventory: before any schema change on a table of nontrivial age, identify who reads it besides the application - export jobs, other services, reporting queries, scheduled dashboards. For Northfield's rename, that inventory turned up an internal reporting export that ran nightly against `orders.buyer_email` directly, owned by a different team on a different release cycle.

Two patterns handle this without forcing every downstream consumer onto the application's timeline:

- **A compatibility view.** Keep a database view (or, during the dual-write window, the old column itself) that presents the old shape, so a consumer's `SELECT` list can go on naming `buyer_email` long after the table stopped having that column. They do still have to repoint the `FROM` clause at the view, which is a one-line change on their side rather than a rewrite - and it has to happen while the old column still exists, not on the day it disappears. The view gets dropped only once every known consumer has confirmed it's no longer needed: not on the application's schedule, but on the slowest consumer's.
- **A deprecation window with an actual notification.** Whoever owns the schema change communicates the timeline to known downstream owners the same way a public API deprecates an endpoint: an announced window, not a silent drop. This only works if the inventory step actually happened - a consumer nobody knew about can't be notified.

Both patterns above are damage control for consumers that already read columns directly. The more durable fix, when the downstream consumer is an HTTP client rather than a direct database reader, is to never let that exposure exist in the first place: don't return `Model::toJson()` or a bare `$order->toArray()` from a controller, and don't let a resource's field names default to mirroring column names. A Laravel API Resource (or a `league/fractal` transformer, on a project that predates or doesn't use Resources) sits between the two and makes the wire format an explicit, independently versioned contract:

```
final class OrderResource extends JsonResource
{
    public function toArray(Request $request): array
    {
        return [
            'id' => $this->id,
            // Wire format is frozen; the column behind it is not.
            'buyer_email' => $this->customer_email,
            'status' => $this->status,
            'total_cents' => $this->total_cents,
        ];
    }
}
```

With that boundary in place, the entire six-phase rename in this chapter can happen without a single HTTP API consumer noticing - the resource is the only place that has to change, and it changes the other way round from what you'd expect: it maps the new column onto the response field the API already promised, instead of the API dictating what the column is allowed to be called.

For the rename, the old `buyer_email` column stayed in place, populated by the dual-write step, until the reporting team confirmed their nightly export had been updated to the new column name - on their schedule, not the application's. Only then did phase 6 ship. Had the export needed more time than the dual-write window allowed, a compatibility view would have covered the gap - shipped during the deprecation window rather than alongside the drop, since a view that appears on the same day the column vanishes has nobody pointed at it yet:

```
-- Ships during the deprecation window, well before phase 6's
drop, so
-- consumers have time to repoint FROM orders at
orders_legacy_shape and
-- keep buyer_email in their SELECT list afterwards.
CREATE VIEW orders_legacy_shape AS
    SELECT id, customer_id, customer_email AS buyer_email, status,
total_cents
    FROM orders;
```

A schema-change checklist

- [] Tested the specific operation (not just "a migration") against a production-sized copy of the table
- [] Confirmed whether the operation takes a blocking lock on the current engine/version
- [] Used a non-blocking index build (`CONCURRENTLY` / `ALGORITHM=INPLACE`) for large tables
- [] Renames/type changes are split into add → dual-write → backfill → migrate reads → drop phases
- [] Backfills run in bounded batches with a pause, not as one statement
- [] Inventoried downstream consumers (exports, other services, BI tools) that read this table directly
- [] HTTP API responses go through a Resource/transformer, not a bare model - column renames don't reach API clients
- [] Downstream consumers notified or bridged with a compatibility view before the old shape is dropped
- [] The "drop" phase is the last, separately reviewed step - never bundled with the rest

Chapter recap

- A migration's cost scales with table size and the specific engine/operation, not with how simple the change looks in code - always test DDL against production-sized data.
- Use non-blocking index builds (`CREATE INDEX CONCURRENTLY` on PostgreSQL, `ALGORITHM=INPLACE`, `LOCK=NONE` on MySQL) on any table large enough that a lock would be noticed.
- Never rename or retype a column in one step - expand it into add, dual-write, backfill, migrate reads, stop dual-write, and drop, each

independently deployable and reversible until the last phase.

- Backfill large tables in bounded, pausable batches (`chunkById` plus a small delay), not a single sweeping `UPDATE`, to protect replication and live traffic.
- Your schema is effectively a public API the moment anything outside your own deploy - an export, another service, a BI tool - reads it directly. Inventory those consumers before assuming a change is safe.
- For consumers that go through HTTP rather than the database directly, a Laravel API Resource (or `league/fractal`) makes the response shape an explicit contract instead of a mirror of column names - the safest fix is the one that prevents the exposure from existing at all.
- Bridge downstream consumers with a compatibility view or an announced deprecation window; let the slowest consumer's timeline, not your migration's convenience, decide when the old shape can actually be dropped.

Chapter 15 - Load Testing & Capacity Planning

The load test that passed, and the launch that didn't

Northfield, a fictional retail platform, is three weeks out from its biggest marketing event of the year - a flash sale expected to drive ten times normal traffic in a two-hour window. The team runs a load test: a script hammers the `/checkout` endpoint with a thousand concurrent requests, latency stays under 200ms, error rate stays near zero. Load test passed. Ship it.

On the day of the sale, the site falls over anyway - not on `/checkout`, but on the *combination* of product-listing traffic, cart updates, and checkout happening together, because real shoppers browse, add to cart, abandon, come back, and only a fraction ever reach checkout at all. The single-endpoint test never exercised that mix, never exercised the database connection pool under concurrent reads-and-writes, and never came close to discovering that the actual constraint that day is the third-party payment gateway's own rate limit, which throttles Northfield's checkout calls once volume passes a threshold nobody had checked.

Nothing about that load test was fake or dishonest. It was just testing the wrong thing: one endpoint, one traffic shape, with no predefined target for what "passing" even meant beyond "didn't 500." This chapter is about closing both gaps - designing a load test that resembles the traffic you'll actually get, and deciding in advance, in the calm of a planning meeting, what numbers count as success.

Define the SLOs before you touch a load-testing tool

A Service Level Objective (SLO) is a target for a Service Level Indicator (SLI) - a specific, measurable signal like "95th percentile response time" or "checkout error rate." Without SLOs written down first, a load test has no way to fail: any result can be read as "fine," because there was never a bar to clear.

Write these down before running anything:

- **Latency targets, as percentiles, not averages.** "p50 < 150ms, p95 < 500ms, p99 < 1500ms" tells you far more than "average response time is 200ms," because an average hides the tail - and the tail is what users and monitoring alerts actually notice. A system with a 200ms average and an 8-second p99 is failing for one in a hundred users on every single request.
- **Error rate ceiling.** "Error rate stays under 0.5% at target load" - and be explicit about what counts as an error: 5xx responses, certainly, but also timeouts, and arguably 429s if the plan didn't intend to be rate-limiting real users at that volume.
- **Throughput target, grounded in a real number.** Not "handle a lot of traffic," but "sustain 2,000 checkout completions per minute for a 90-minute window," derived from the actual marketing forecast (expected visitors x historical conversion rate), not a round number that sounded safe in a meeting.

These three - latency, error rate, throughput - are the core of almost every load test's pass/fail criteria, and writing them down first forces the useful question: *what will we do if we hit the target load and any one of these fails?* If nobody can answer that in the planning meeting, the load test is going to

surface a decision nobody's ready to make, under worse time pressure, on the day it matters.

Traffic shape: model the journey, not the endpoint

Real traffic is not N identical requests to one URL. A realistic load test reproduces the *mix* and *sequencing* of what users actually do, because bottlenecks frequently live in the interaction between paths, not any single path in isolation.

For the flash-sale scenario, a defensible traffic model looks like a weighted set of user journeys rather than a single script:

```
{
  "journeys": [
    { "name": "browse_only", "weight": 55, "steps": ["home",
"category", "product"] },
    { "name": "browse_and_abandon_cart", "weight": 25, "steps":
["product", "add_to_cart", "cart"] },
    { "name": "full_checkout", "weight": 15, "steps": ["product",
"add_to_cart", "cart", "checkout", "payment"] },
    { "name": "returning_search", "weight": 5, "steps": ["search",
"product", "add_to_cart", "checkout", "payment"] }
  ],
  "think_time_ms": { "min": 500, "max": 4000 },
  "full_checkout_ramp": {
    "from_completions_per_minute": 50,
    "to_completions_per_minute": 2000,
    "over_minutes": 15
  }
}
```

Two details matter as much as the journey mix itself. First, **think time** - the pause between a real user's clicks - matters because it changes how many requests are in flight concurrently versus how many total requests land over a window; a script with zero think time can generate the right total volume while wildly overstating concurrent connection pressure (or understating it, depending on what you're testing for). Second, the **ramp**, not just the peak - a flash sale doesn't teleport to 2,000 checkout completions per minute, and a system that survives a 15-minute ramp to peak can still fail against a 90-second spike to the same peak, because caches haven't warmed and autoscaling hasn't caught up. If spiky arrival is plausible for your event, test that shape specifically (see the load-test type table below) rather than

assuming a gentle ramp covers it.

The `full_checkout` journey above turns directly into a k6 scenario. k6 (from Grafana Labs) is the tool used throughout this chapter's examples: a load-testing tool where the test script itself is JavaScript, run from the command line via `k6 run checkout.js`. The executor choice matters more than it looks. The target you wrote down was 2,000 completed checkouts per minute, so reach for `ramping-arrival-rate` rather than the more familiar virtual-user stages - VU stages hold a number of concurrent workers, and the rate they actually produce falls out of how fast your app responds, which means the test quietly eases off exactly when the system starts struggling. Arrival rate schedules the iterations regardless:

```
import http from 'k6/http';
import { sleep } from 'k6';

export const options = {
  scenarios: {
    full_checkout: {
      executor: 'ramping-arrival-rate',
      timeUnit: '1m',
      startRate: 50, // the baseline from the traffic model above
      preAllocatedVUs: 300,
      maxVUs: 500,
      stages: [
        { duration: '15m', target: 2000 }, // checkout completions
        { duration: '90m', target: 2000 }, // hold for the sale
      ],
    },
  },
};
```

```
    },  
    thresholds: {  
      http_req_duration: ['p(95)<500', 'p(99)<1500'],  
      http_req_failed: ['rate<0.005'],  
    },  
  },  
};  
  
export default function () {  
  http.get('https://shop.example.com/product/1234');  
  sleep(Math.random() * 3.5 + 0.5); // think time, matching  
  think_time_ms above  
  http.post('https://shop.example.com/cart/add', { product_id:  
1234 });  
  sleep(Math.random() * 3.5 + 0.5);  
  http.get('https://shop.example.com/checkout');  
  sleep(Math.random() * 3.5 + 0.5);  
  http.post('https://shop.example.com/payment', { method: 'card'  
});  
}
```

Each of the other journeys in the model would get its own scenario, weighted to its share of the mix.

Four kinds of load test, and what each one is actually for

Test type	What it does	What it answers
Load test	Sustained traffic at (or near) expected peak, realistic mix, realistic ramp	"Do we meet our SLOs at the volume we expect?"
Stress test	Push volume past expected peak until something breaks	"Where's the ceiling, and what breaks first when we exceed it?"
Soak test	Moderate, sustained load over hours, not minutes	"Do we leak memory, connections, or disk over time under continuous load?"
Spike test	Sudden jump from baseline to peak with no ramp	"Does autoscaling/caching keep up with instantaneous demand, or is there a painful cold-start window?"

A pre-launch checklist for a known high-traffic event usually wants all four, in that order: load first (does the expected case work at all), stress second (how much headroom exists above the forecast, in case the forecast is wrong - and it usually is), soak third (does the system degrade over the multi-hour window the event actually runs for), and spike last (does the traffic curve's steepest section, not just its peak, hold up).

Finding the real bottleneck instead of guessing at it

A load test's most valuable output usually isn't the pass/fail verdict, it's the trail of evidence for *where* the system degraded first. That requires instrumenting every layer the request touches during the test, not just watching the application's own response time:

- **Database.** Connection pool saturation, slow query log entries that only appear under concurrency (lock waits, not just slow individual queries), replica lag if reads are offloaded.
- **Queue.** Job backlog depth over time - a queue that's "keeping up" at low load can start accumulating a backlog the moment throughput crosses the workers' effective processing rate, and a backlog that only grows never recovers on its own (see Chapter 11).
- **Cache.** Hit rate under the test's actual access pattern - a synthetic test that hits the same ten product IDs repeatedly will show a hit rate no real, long-tail catalog traffic will ever produce.
- **External dependencies.** Every third-party call the request path makes, with its own latency and error tracking. This is where the flash-sale scenario's real answer lives: the application layer holds up fine at ten times normal traffic, but the payment gateway's own published rate limit is lower than the target checkout throughput, so a fraction of legitimate checkouts start getting rejected by the gateway itself, not by anything in the platform's own control. No amount of application-side tuning fixes that; the fix is negotiating a higher limit with the vendor, queuing/smoothing checkout submissions client-side, or accepting a lower throughput ceiling and communicating it to the business ahead of time - all decisions that need to be made weeks before the event, not discovered during it.

The general principle: **load-test with the same observability you'd want during a real incident.** If a metric isn't visible during the test, the test can only tell you *that* something broke, not *why* - and "why" is the only actionable output.

From test results to a capacity plan

A load test tells you what happened at the load you tested. A capacity plan turns that into: how much load can we actually take, how do we know when we're approaching it, and what do we do next.

- **Headroom, not just pass/fail.** If SLOs hold at 2,000 requests per second, push further in the stress test until they don't, and record that ceiling. "We pass at target load" and "we pass at target load with 3x headroom before failure" imply very different risk postures for the actual event.
- **Leading indicators, not lagging ones.** Translate the bottleneck found during testing into a dashboard metric and an alert threshold that fires *before* users are affected - queue depth crossing a threshold, connection pool utilization crossing 80%, gateway error rate ticking up - so the response happens during the ramp, not after the SLO has already been breached.
- **A pre-agreed response plan.** Decide, in advance, what happens if a leading indicator fires during the real event: is there a feature flag to disable a non-critical path, a manual scaling trigger, a fallback for the constrained dependency? Writing this down before the event turns a live incident into "run the plan" instead of "improvise under pressure."
- **Revisit before the next event, not after.** Capacity plans rot. The catalog grows, the query patterns shift, a new feature added a database

call to the checkout path since the last test. Treat load testing as a recurring practice ahead of every high-stakes traffic event, not a one-time certification.

A leading-indicator check like the queue-depth example above might run as a scheduled job, translating the threshold into something that actually fires:

```
/*
 * Runs on a schedule, checking the bottleneck found during load
 * testing.
 */
$queueDepth = Queue::size('checkout');

if (
    $queueDepth > 5000 && ! Cache::has(
        'checkout-queue-alert-cooldown'
    )
) {
    Notification::route('slack', config('alerts.onscall_webhook'))
        →notify(new QueueDepthExceeded('checkout', $queueDepth));

    Cache::put(
        'checkout-queue-alert-cooldown',
        true,
        now()→addMinutes(10)
    );
}
```

The threshold and cooldown are tuned per dependency, but the shape - check, compare, notify once - stays the same.

Chapter recap

- Write SLOs (latency percentiles, error rate ceiling, throughput target grounded in a real forecast) before running a load test, or "passing" has no meaning.
- Model realistic traffic: a weighted mix of user journeys with think time and a ramp that matches the expected arrival curve, not a single endpoint hammered at constant concurrency.
- Load, stress, soak, and spike tests answer different questions - use the one that matches the risk you're actually worried about, and run all four ahead of a genuinely high-stakes event.
- Instrument every layer the request touches (database, queue, cache, external dependencies) during the test, because the real bottleneck - like a third-party payment gateway's own rate limit - is often outside the application layer entirely and invisible if you only watch your own response times.
- Turn results into leading-indicator alerts and a pre-agreed response plan, and re-run the test before the next high-traffic event, not just the first one.

